Multimodal CTPA Triage Workflow: Deployment Readiness Evaluation

Prepared for the Deployment Review Committee | Healthcare AI Coordination Analyst, Regional Radiology Network | 2 November 2026

Synthetic training material—not clinical guidance. No live Microsoft Foundry access is required.

Evidence base. Workflow diagram (cited as Diagram, Step n); ML Performance Summary (ML Tables 1–4); Deep Learning Output Comparisons (DL-01 to DL-04); Computer Vision Output Set (CV-01 to CV-06). The Scenario Guide was not available for this review, so no criteria are drawn from it; every threshold and target below is quoted from the supplied materials.

> **Recommendation: proceed only with mitigations.** The model meets all three scenario acceptance criteria at threshold 0.58 but fails the low-contrast subgroup minimum. Five conditions (M1–M5, Section 2) must be in place at go-live.

Section 1—Workflow Overview

One-sentence summary: The workflow joins imaging, clinical text and patient context at a single linkage checkpoint, uses AI only to extract context and score studies for queue priority, and leaves every interpretation to a radiologist who can override the AI's priority.

- **Assumption:** Decision 6 does not say which branch wins when a study has both a high score and a quality or context flag, so the flag is assumed to take precedence: technical-quality review first, then the priority queue if the score is ≥ 0.58.

How the data connects

Three inputs enter in parallel (Diagram, Steps 1A–1C): the synthetic CTPA imaging study with its Study UID, accession number and acquisition time; the clinical text, made up of the order indication and a prior-report excerpt tied to an order ID and encounter token; and de-identified patient context, an age band and a prior-study flag. Step 2 joins them on three keys: encounter token, order ID and Study UID. If any key is missing, automated routing stops. This is the control that prevents a score from being attached to the wrong patient's history (Diagram, Step 2). Step 3 normalises metadata, sets a contrast-quality flag and checks that required context is present, with the package due within 5 minutes of imaging arrival. Text and image then go to two AI steps in parallel (Steps 4A, 4B), and Step 5 combines the triage score, extracted context and quality flags into a routing input. Step 8 audits latency, discordance and subgroup performance and feeds back to the linkage step and to the data and model teams.

Table 1. Automated, AI-assisted and human steps (Diagram)

Step	Who acts	What it may do	Trust boundary
2 Linkage validation	Automated rule	Join the three linkage keys	Missing key stops automated routing
3 Preparation/quality gates	Automated rule	Normalise metadata; set contrast-quality flag; check context	Must finish ≤ 5 min after imaging arrival
4A NLP context extraction	AI-assisted	Extract indication and prior-history cues	No performance evidence supplied (Section 2)
4B Imaging model	AI-assisted	Critical-triage score; optional regional cue	Prioritises only; cue is not a diagnosis

Step	Who acts	What it may do	Trust boundary
5–6 Review logic and routing	Automated rule	Combine score, context and flags; select a queue	No autonomous diagnosis or clearance
7A Priority radiologist review	Human (radiologist)	Read original study ≤ 30 min	May override AI priority
7B Standard radiologist review	Human (radiologist)	Final interpretation ≤ 4 h	Routine clinical queue
7C Technical quality review	Human (technical)	Resolve quality/context problem ≤ 60 min	Returns study to radiologist workflow
8 Monitoring/audit	Human-governed	Track latency, discordance, subgroups	Feeds back to data and model teams

Constraints the committee should note

- **Latency:** package ≤ 5 min after imaging arrival; priority review ≤ 30 min; technical review ≤ 60 min; standard review ≤ 4 h (Diagram, Steps 3, 7A–7C; CV destinations table). No measured latency was supplied, so these are targets, not observed performance.
- **Essential inputs:** all three linkage keys; the order indication and prior-report excerpt for Step 4A; the age band and prior-study flag; and the contrast-quality flag. Missing context routes to technical-quality review, not to the standard queue (Diagram, Decision 6).
- **Override points:** the radiologist reads the original study in every path and may override priority (Diagram, Step 7A). There is no autonomous clearance path (CV destinations table).
- **Open design gap:** the diagram says a failed linkage stops automated routing but does not say where the study goes next. The network must define a manual hand-off to a radiologist queue before go-live.

Section 2—ML Output Evaluation

One-sentence summary: At threshold 0.58 the model meets every scenario acceptance criterion overall but misses the low-contrast subgroup minimum, so it is ready only for a rollout with mitigations that send low-contrast studies to technical-quality review whatever their score.

Table 2. Performance at threshold 0.58 against scenario criteria (ML Tables 1, 2, 4)

Metric	Result	Scenario criterion	Status
Sensitivity (critical class)	0.960 (288/300)	≥ 0.95	**Met**
False-negative rate	0.040 (12/300)	≤ 0.05	**Met**
Precision/PPV (critical class)	0.593 (288/486)	≥ 0.58	**Met, margin 0.013**
Sensitivity, adequate contrast	0.980 (245/250)	≥ 0.90	**Met**
Sensitivity, borderline/low contrast	0.860 (43/50)	≥ 0.90	**Not met**
Specificity	0.780 (702/900)	None set	Cannot assess
F1 / accuracy / ROC-AUC	0.733 / 0.825 / 0.940 (0.920–0.960)	None set	Reference only

What the numbers mean operationally

Of 300 reference-critical studies, 288 reach the priority queue and 12 do not (ML Table 2). Without a quality gate, those 12 would wait in the 4-hour standard queue. The priority queue receives 486 of 1,200 studies (40.5%), and 198 of them, about two in five, are reference-non-critical. That is the false-positive load the ≤ 30-minute target must absorb. Precision clears its floor by only 0.013 and was measured on a test set in which 25% of studies were critical (300/1,200). Precision moves with case mix, so it must be re-measured on live volume.

Changing the threshold does not solve these problems. Of the three supplied thresholds, only 0.58 meets all three criteria: 0.45 drops precision to 0.505, and 0.70 drops sensitivity to 0.910 and raises the false-negative rate to 0.090 (ML Table 3). No subgroup results are supplied for 0.45 or 0.70, so it is unknown whether any threshold would close the low-contrast gap.

Low-contrast subgroup

Low-contrast studies are 17% of critical cases (50/300) but account for 7 of the 12 misses (58%). Their miss rate is 14% (7/50) against 2% (5/250) for adequate contrast, and their sensitivity of 0.860 is below the scenario minimum of 0.900 (ML Table 4). The workflow already contains a mitigation: the Step 3 contrast-quality flag routes affected studies to technical-quality review (≤ 60 min), which returns them to a radiologist (Diagram, Steps 3, 6, 7C). That mitigation works only if the flag catches low-contrast studies, and no evidence on the flag's accuracy was supplied.

Missing evidence

Readiness cannot be determined for the following, because the materials supply no acceptance criterion or no data:

- **Specificity and false-positive burden:** no minimum is set and no priority-queue capacity data is supplied, so it cannot be judged whether 198 false priorities per 1,200 studies is sustainable at ≤ 30 minutes.
- **Clinical-text extraction (Step 4A):** no metrics or criteria are supplied, so the text component's readiness is undetermined.
- **Linkage and patient context:** no linkage-failure rate and no performance by age band or prior-study flag.
- **Regional-cue localisation:** no overlap or offset criterion exists (Section 3).
- **Latency and contrast-flag accuracy:** targets are stated, but neither has been measured.

Recommendation: proceed only with mitigations

A controlled rollout without conditions is not supported, because one scenario criterion is failed (ML Table 4) and the regional cues show failure modes (Section 3). Delay and retraining are not required, because all three headline criteria are met (ML Table 1), the workflow already has a route that covers the subgroup gap, and every study is read by a radiologist. Go-live should require:

- **M1:** Send every contrast-flagged study to technical-quality review whatever its score, then to the priority queue whatever its score, until low-contrast sensitivity reaches 0.90. A score below threshold is not reliable in this subgroup (ML Table 4; CV-04).
- **M2:** Present regional cues as attention aids labelled non-diagnostic, with an artifact warning when a quality flag is set (DL-03, DL-04).

- **M3:** Allow Step 4A text cues to raise priority but not lower it until text-extraction evidence exists.
- **M4:** At Step 8, track sensitivity, false-negative rate, precision and latency by contrast subgroup. Any result outside the scenario criteria (sensitivity < 0.95, false-negative rate > 0.05 or precision < 0.58) triggers committee review.
- **M5:** Define the manual hand-off for linkage failures before go-live.

Section 3—Deep Learning Failure Mode Review

One-sentence summary: The regional cue fails independently of the triage score: all four paired cases scored above 0.58, yet in three the cue was undersized, displaced or placed where no reference lesion is marked (DL-02, DL-03, DL-04), so cues cannot be treated as lesion boundaries.

DL-01 is the comparison case: in an adequate-contrast study the cue closely matches the reference (overlap 0.79, area ratio 0.86, centroid offset 2.2). Each of the other three cases shows a different failure, and the artifact pattern recurs in the computer-vision set. Each pattern rests on one paired example, so the materials show what can go wrong but not how often; frequency must come from validation.

Table 3. Failure modes, risk and follow-up

Failure mode	Evidence	Type and operational risk	Immediate mitigation	Longer-term action
Under-segmentation	DL-02: overlap 0.31, area ratio 0.31, centroid offset 0.0; borderline contrast; score 0.63	Extent error with correct position. Moderate risk: triage is unaffected (score ≥ 0.58), but the cue covers about a third of the reference region. Occurs in the contrast subgroup that fails its minimum (ML Table 4); this is an association, not a proven cause.	Show cues as markers, not boundaries; display the contrast flag with the cue (M2).	Add pixel-level labels for borderline/low-contrast studies and contrast-varied augmentation; report overlap by contrast subgroup.
Localisation drift	DL-03: overlap 0.01, centroid offset 29.5 (units not stated), area ratio 0.78; adequate contrast; score 0.71	Wrong-location error. Highest cue risk: a plausibly sized cue points away from the reference region, and because contrast is adequate the quality gate will not catch it.	Radiologist reads the full original study regardless of cue (CV destinations table); log cue–reference disagreements at Step 8 (M4).	Focused localisation validation study with a pre-agreed overlap and offset criterion, which does not yet exist.
Artifact-associated false cue	DL-04: no reference lesion, overlap 0.0, motion artifact; score 0.66. Recurs in CV-03 (motion) and CV-05 (streak)	False-positive detection co-located with artifact. Moderate risk: no missed finding, but it adds unnecessary priority-queue load and a misleading cue.	Artifact flag routes to technical-quality review (M1 path); cue carries an artifact warning (M2).	Augment training with motion and streak artifacts, including artifact-only negative studies; report the false-cue rate by artifact type.

Remediation priority: fix localisation drift first. It is the only failure the quality gate cannot intercept. The two quality-associated failures are already contained by the routing mitigations in Section 2.

Section 4—Computer Vision Output Assessment

One-sentence summary: Half the outputs carry image-quality problems that make the score unreliable, most seriously CV-04, a critical study scored 0.41 that score-only routing would send to the 4-hour queue, so routing must apply the quality flag before the score.

Routing uses only what is known at triage time: score, output type and acquisition-quality note. Reference adjudication is used only to assess review status, because it is not available when a study is routed.

Table 4. Output type, review status and routing (CV-01 to CV-06)

Case	Score / quality	Output type	Review status	Routing (target)	Supporting evidence
CV-01	0.88 / adequate	Image-level finding	No discrepancy indicated	Priority radiologist (≤ 30 min)	Score above 0.58 agrees with critical reference; no quality flag.
CV-02	0.81 / adequate	Regional cue	No discrepancy indicated (image level)	Priority radiologist (≤ 30 min)	Score agrees with critical reference. No reference region is drawn, so the cue's position is unverified (compare DL-03).
CV-03	0.67 / motion artifact	Regional cue	Suspected artifact-related misread	Technical quality (≤ 60 min), then priority radiologist	Above-threshold score on a non-critical study; the cue sits over motion bands, as in DL-04.
CV-04	0.41 / borderline-low contrast	Image-level finding	Suspected artifact-related misread (acquisition quality)	Technical quality (≤ 60 min), then priority radiologist (M1)	Critical study scored below threshold; low contrast is the subgroup below its minimum (ML Table 4).
CV-05	0.72 / severe streak artifact	Regional cue	Indeterminate	Technical quality (≤ 60 min), then priority radiologist	Reference review is itself indeterminate, so a misread cannot be confirmed; the cue box overlies streak lines.
CV-06	0.18 / adequate	Image-level finding	No discrepancy indicated	Standard radiologist (≤ 4 h)	Low score agrees with non-critical reference. Still read by a radiologist: there is no autonomous clearance.

Why the escalated cases need review

CV-01 and CV-02 go to priority review because a score above 0.58 on an adequate-quality study is where the model is most reliable (adequate-contrast sensitivity 0.980, ML Table 4), yet the score still does not establish a diagnosis (CV destinations table). CV-03, CV-04 and CV-05 go to technical-quality review because their scores are least trustworthy in exactly these conditions: CV-04 is the kind of false negative that ML Table 4 predicts, and CV-03 and CV-05 repeat the artifact-associated cue seen in DL-04. All three then return to the priority queue, because CV-03 and CV-05 score above 0.58 and CV-04 falls under M1.

Routing procedure for flagged cases

1. Confirm linkage and required context (Steps 2–3). If a key is missing, stop automated routing and hand the study to a radiologist queue manually (M5).
2. Apply quality and context flags before the score. Any contrast, motion or streak flag, or missing context, goes to technical-quality review (≤ 60 min) whatever the score.
3. After technical review, return the study to the priority queue if it scored ≥ 0.58 or carried a contrast flag (M1); otherwise, return it to standard review.
4. For unflagged studies, send scores ≥ 0.58 or score–context discordance to priority review (≤ 30 min) and all others to standard review (≤ 4 h).
5. Show regional cues as non-diagnostic attention aids, with an artifact warning on flagged studies (M2).
6. The radiologist reads the original study in every path and may override priority; overrides and cue disagreements are logged to Step 8 (M4).